

Digital Systems Lab – Assignment 1

INSTRUCTIONS:

- Install Xilinx Vivado software.
- Write Verilog design code and testbench.
- Simulate the design and verify output.
- Take screenshots of code and waveform.
- Paste screenshots in documentation.

QUESTION 1: Combine Two 4-bit Numbers

Write a Verilog module that combines two 4-bit inputs into one 8-bit output.

Inputs: a[3:0], b[3:0]

Output: y[7:0]

Upper 4 bits = a, Lower 4 bits = b

QUESTION 2: Extract Lower Nibble

Write a Verilog module that outputs the lower 4 bits of an 8-bit input.

Input: in[7:0]

Output: out[3:0]

QUESTION 3: Bitwise AND or OR Selector

If sel = 0 → output = a & b

If sel = 1 → output = a | b

Inputs: a[3:0], b[3:0], sel

Output: y[3:0]

QUESTION 4: Simple Even Parity Generator

Generate even parity for a 4-bit input.

Input: data[3:0]

Output: parity

Parity = 1 if number of 1s is odd.

QUESTION 5: Swap Two Bits

Given a 4-bit input, swap bit0 and bit3.

Input: in[3:0]

Output: out[3:0]

Bits 1 and 2 remain unchanged.